

Confidence Interval for σ^2 :

Thus far, we have found confidence interval around various other estimators.

Recall that $\frac{(n-1)s^2}{\sigma^2}$ follows χ^2

distribution with $(n-1)$ df.

$$\Pr\left(a \leq \frac{(n-1)s^2}{\sigma^2} \leq b\right) = 0.9$$

An experimenter wanted to check the variability of measurements obtained by using equipment designed to measure the volume of an audio source. Three independent measurements recorded by this equipment for the same sound were 4.1, 5.2, and 10.2. Estimate σ^2 with confidence coefficient 0.90.